

Abdelrahman Essam

Software Engineering Student | Full-Stack & Systems Developer

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Summary

Computer & Communication Engineering student (GPA 3.76/4.0, top 20 in cohort) applying competitive-programming problem-solving to scalable full-stack systems — shipping and **load-testing** production apps with Java, Spring Boot, React/Next.js, NestJS, PostgreSQL, MySQL, Kafka, and Docker. Seeking a software engineering internship or junior developer role.

Education

Faculty of Engineering, Alexandria University

Oct 2022 – July 2027

B.Sc. in Computer and Communication Engineering

Alexandria, Egypt

– **GPA: 3.76/4.0** — Ranked in the top 20 of the cohort.

– **Relevant Coursework:** Algorithms, Data Structures, OS, Computer Networks, Databases, Software Engineering, OOP, Multithreading, Design Patterns.

Technical Skills

Languages: Java, C++, C, Python, JavaScript, TypeScript, Bash, C# **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend: Spring Boot, Spring Security, NestJS, Node.js, .NET, REST APIs, JWT, JPA/Hibernate, Microservices

Databases: MySQL, PostgreSQL, SQL, Relational Database Design, Query Optimization

DevOps & Tools: Docker, AWS (EC2, RDS), Nginx, Linux, Apache Kafka, Zookeeper, pgBoss, Git, GitHub Actions, CI/CD, Automated Testing, Maven, JUnit, Mockito, JaCoCo, Jest, JMeter

Core CS: Data Structures, Algorithms, OOP, Design Patterns, Multithreading, POSIX, RBAC, WebSockets

Experience

Mylo | finX Squad | *React, Next.js, NestJS, TypeScript, PostgreSQL, pgBoss*

Aug 2026 – Present

Full-Stack Software Engineering Intern

Egypt

– Shipped to **production** a **React/Next.js** loan rescheduling view for handling **multiple loans per consumer**.

– Cut average handling time per request from **15 to 9 minutes**.

– Fixed a **NestJS** metadata bug that dropped loan updates from the **LMS** service; restored processing for **100%** of affected requests.

– Re-architected the flow around **pgBoss (PostgreSQL)** job queues in production: **500+ jobs/day**, zero dropped jobs.

– Migrated **10 UI components** to **MyloKit** and **3 BFF endpoints** to **Next.js Server Actions**, removing an intermediate API layer.

EJADA Systems | GitHub | *Java, SQL, Spring Boot*

Jul 2026 – Aug 2026

Software Engineering Intern

Egypt

– Designed relational schemas and **20+ complex SQL queries** for enterprise-style data models.

– Built Java modules exercising **multithreading** and **exception handling** under Agile/Scrum code review.

Projects

RentSphere — Rental Platform | GitHub | *Spring Boot, React.js, MySQL, AWS, Docker, JMeter*

Oct 2025 – Jan 2026

– Built a full-stack rental marketplace with **27 RESTful APIs** and **3-role Spring Security RBAC (JWT, BCrypt)**.

– Implemented property search/filter, rental requests, contracts, favorites, and **PayPal** installment checkout.

– Modeled **8 MySQL tables**; tuned indexing **18 → 12** with **EXPLAIN**-verified query paths on a seeded **100k-listing** dataset.

– **JMeter** tests (25–200 users) exposed **N+1** image queries; batched via chunked **IN** fetches — mean latency **1.7 s → 42 ms** at 50 users.

– Sustains **200 concurrent users** at **0% errors** (≈ 71 req/s); **p95** ≈ 0.3 s at 100 users, detail **p95** ≈ 15 ms on 100k rows.

– Deployed a **4-service Docker Compose** stack (**MySQL 8, Spring Boot, React, Nginx**) on **AWS EC2 + RDS** with **rate limiting** and **Actuator** metrics.

– CI/CD via **GitHub Actions**: **233 automated tests** (**182 JUnit/Mockito, 51 Jest**), **JaCoCo** 72% coverage, Docker build gates.

IoT Weather Station Monitoring | GitHub | *Java, Apache Kafka, Zookeeper, Docker*

Feb 2026 – May 2026

– Engineered a concurrent acquisition system in **Java** ingesting 10 IoT stations via multi-threaded executor pools.

– Streamed live **Open-Meteo** data into **Apache Kafka** (**acks=all**, idempotent) via a REST adapter; multi-stage **Docker** builds.

Social Media Application | GitHub | *Spring Boot, React.js, MySQL, JWT, WebSockets, Apache JMeter*

May 2025 – Sep 2025

– Produced **45+ RESTful APIs** with role-based, stateless **JWT** security.

– Implemented real-time messaging with **WebSockets** (STOMP) for 100+ concurrent users.

– Cut average response time from **80 ms to 35 ms** via **JPA fetch optimization**, validated with **JMeter**.

Achievements

▪ **ECPC 2025 Finalist** — ranked **98th nationally**; **1,000+** problems on LeetCode & Codeforces (graphs, DP, system design).